Partially Ordered Stochastic Conformance Checking: A Review

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1 The Technique

Leemans et al. [1] generalise Earth Movers' Stochastic Conformance (EMSC) from totally ordered to partially ordered traces. Their approach is based on a semantic view. When two activities are executed concurrently, the order in which they are recorded does not carry any process information. Therefore, any conformance method that depends on this order is measuring process properties rather than the properties themselves. The technique therefore re-bases conformance checking on a different semantic object — the *stochastic partially ordered (PO) language* — and uses EMSC to measure the stochastic distance at that level. Three concrete limitations of standard EMSC are addressed simultaneously: **P1**: uncertain event order in logs (i.e. events sharing a timestamp); **P2**: redundant assignment of probability to interleavings of concurrent behaviour; and **P3**: the absence of any formal guarantee on values returned for models containing loops.

1.1 Semantic idea of partially ordered (PO) traces

A PO trace is a triple $\rho = (E, \prec, l) \in \mathcal{P}$ with finite event set E, strict partial order $\prec \subseteq E \times E$, and labelling $l \colon E \to \mathcal{A}$. Its set of linearisations $\mathcal{L}(\rho) \subseteq \mathcal{A}^*$ collects every totally-ordered trace consistent with $\prec$, so one PO trace stands for a family of TO-traces. As a simple example, consider the two traces $\langle a, b, c\rangle$ and $\langle a, c, b\rangle$. These would be consolidated to a trace which would be seen as $\langle a, \{b, c\}\rangle$ indicating that the concurrent events b and c can occur in either order. A *stochastic PO language* is a map $L \colon \mathcal{P} \to [0, 1]$ with $\sum_\rho L(\rho) = 1$, summing over all PO traces with concurrent interleavings. PO traces provide two interpretations: *certain*, where the events can be executed in any order (used for models, addressing **P2**), and *uncertain*, where one linearisation was realised but cannot be recovered from the data (used for logs with imprecise timestamps, addressing **P1**).

1.2 Embedding Models and Logs into the Semantics

The model side is a labelled stochastic Petri net (LSPN) $PN = (P, T, F, \Sigma, \lambda, M_0, w)$ with weight function $w \colon T \to \mathbb{R}^+$ assigning probabilities. The authors require soundness, safeness and *confusion-freeness*: any two transitions t_1, t_2 with $\bullet t_1 \cap$

$\bullet t_2 \neq \emptyset$ and $\bullet t_1 \neq \bullet t_2$ are never jointly enabled. Under these properties the probability of a run (trace) r depends only on transitions involved in choices; weights on concurrently firing transitions cancel out,

$$PN(r) \;=\; \prod_{t' \in T'} \frac{w(\beta(t'))}{\sum_{t \in T \,\mid\, \bullet t \,=\, \bullet \beta(t')} w(t)}, \tag{1}$$

so the order in which concurrent transitions are interleaved leaves $PN(r)$ unchanged. Algorithm 1 [1] generates the language of partially ordered traces and their probabilities. It does so by performing a recursive play-out, which fires maximal sets of pairwise-independent (concurrent) enabled transitions in parallel at each step. Each run defines a PO trace by removing silent transitions and all places while keeping the partial order over the remaining events; summing $PN(r)$ over runs sharing a PO trace yields L_M(probabilities of traces discovered by the LSPN). Because loops lead to infinitely many runs, the play-out is *truncated* after q most-likely plus r random runs. The remaining probability, which stays unconsidered, is the residual mass $L_M(\perp)$, which is treated as an extra symbolic PO trace, addressing **P3**.

On the log side, each case represents one PO trace by reading the $\prec$-relation off the timestamps—events sharing a timestamp stay incomparable—and empirical frequencies are normalised to L_L under the uncertain interpretation; reliable timestamps allow the certain interpretation instead.

1.3 Distance, EMSC, and Bounds

A trace–trace distance δ (the paper uses normalised Levenshtein) is lifted to a PO-trace distance Δ. The normalised Levenshtein distance is similar to the concept of alignments but adds a fourth move type called "substitution" denoting two non-matching events in both traces. Because each side may be certain or uncertain, depending on the interpretation of model traces (L), four variants of interpretation arise: Δ_δ^{cc} (both certain, minimum over pairs of linearisations), $\Delta^{uu\text{-best}}/\Delta^{uu\text{-worst}}$ (both uncertain), and mixed minimax $\Delta^{uc\text{-best}}/\Delta^{uc\text{-worst}}$. EMSC is then the standard optimal-transport problem

$$\mathrm{emsc}_\Delta(L, L') \;=\; 1 \;-\; \min_R \sum_{\rho \in \tilde{L}, \rho' \in \tilde{L}'} R(\rho, \rho')\, \Delta(\rho, \rho'), \tag{2}$$

revise the following paragraph

subject to the marginal constraints $L(\rho) = \sum_{\rho'} R(\rho, \rho')$ and $L'(\rho') = \sum_{\rho} R(\rho, \rho')$, but now indexed over PO traces. Δ^{cc} is computed by an A^* search over a four-move alignment (synchronous, substitution, top, bottom), with an admissible heuristic equal to the multiset difference of yet-unaligned labels. When L_M is truncated and/or L_L is uncertain, the true EMSC is bounded: absorbing $L_M(\perp)$ into the symbolic trace $\perp$ with cost 0 (lower bound) or 1 (upper bound) and selecting the appropriate Δ-variant yields an interval $[\mathrm{emsc}_\uparrow, \mathrm{emsc}_\downarrow]$ that provably contains the true value (Lemmas 4 and 5). The gap is equal to $1 - \sum_{\rho \in \tilde{L}_M} L_M(\rho)$ and shrinks monotonically as q and r grow.

2 An Example Exercising P1, P2, and P3

Consider a simple loan application process, which starts with applying for a loan and ends with either rejection or approval. For this process, I have created a synthetic event log L_1 (Table 1) and a perfectly fitting Petri net model (Figure 1). The labelled stochastic Petri net in Figure 1 captures all three problems: *Verify Credit* (VC) and *Verify Income* (VI) are concurrent (**P2**), the *Revise* (Rv) loop admits runs of unbounded length (**P3**), and in the log the two verifications share a timestamp, so their order is uncertain (**P1**). The net is sound, safe, and confusion-free, as the technique demands. The stochastics were not included in the figure, but considered in the evaluation of the method, where the model's stochastic PO language L_M is discovered from the original log L_1 and compared to the original log and two further logs with different frequency distributions.

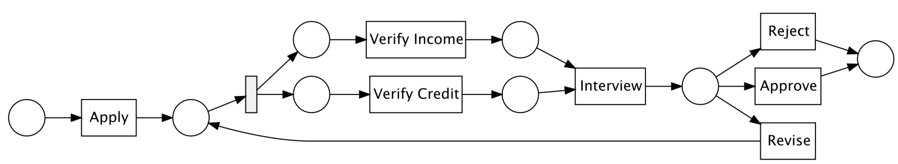

Fig. 1. A perfectly fitting Petri net with the underlying log.

The model's stochastic PO language L_M is discovered from the original log L_1, so it matches L_1 by construction. The earth mover's stochastic conformance (EMSC) distance calculation will be executed against L_1 and two further logs over the *same nine variants* but with different frequency distributions: a variation log L_2 and a strongly loop-biased log L_3 (Table 1). Through the concurrency of VC and VI the two interleavings are linearisations of one PO trace, so the nine variants collapse into the five PO traces $\rho_1, \ldots, \rho_5$, on which L_1 places mass $(60, 30, 6, 3, 1)$, L_2 places $(55, 20, 11, 10, 4)$, and L_3 places $(4, 30, 58, 3, 5)$. While L_2 mildly shifts mass from the short paths toward the loop, L_3 concentrates it almost entirely on the looping trace ρ_3.

Using the Ebi CLI, I have sampled the model's partially ordered traces (`ebi sample partially-ordered-traces`, 1000 runs) and computed EMSC against each log under the uncertain interpretation. L_1 scored 0.996; the missing 0.004 is the residual mass $L_M(\perp)$ left by truncating the loop (**P3**). L_2 scored 0.926 and L_3 only 0.744, since the transport must move the loop-heavy masses onto the short traces L_M favours: the more a log deviates from L_M, the more mass travels at a positive PO-trace distance. The ordering $\text{emsc}(L_M, L_3) = 0.744 < 0.926 = \text{emsc}(L_M, L_2) < 0.996 = \text{emsc}(L_M, L_1)$ confirms that L_3 is stochastically furthest from the model.

Trace variant	**PO trace**	$\mathbf{L_1}$	$\mathbf{L_2}$	$\mathbf{L_3}$
$\langle$A, VC, VI, I, Ap$\rangle$	ρ_1	30	15	1
$\langle$A, VI, VC, I, Ap$\rangle$	ρ_1	30	40	3
$\langle$A, VC, VI, I, R$\rangle$	ρ_2	16	10	1
$\langle$A, VI, VC, I, R$\rangle$	ρ_2	14	10	29
$\langle$A, VC, VI, I, Rv, VC, VI, I, Ap$\rangle$	ρ_3	3	6	1
$\langle$A, VI, VC, I, Rv, VI, VC, I, Ap$\rangle$	ρ_3	3	5	57
$\langle$A, VC, VI, I, Rv, VC, VI, I, R$\rangle$	ρ_4	2	5	2
$\langle$A, VI, VC, I, Rv, VI, VC, I, R$\rangle$	ρ_4	1	5	1
$\langle$A, VI, VC, I, Rv, VI, VC, I, Rv, VI, VC, I, Ap$\rangle$	ρ_5	1	4	5
Total		**100**	**100**	**100**

Table 1. Variant frequencies of the original log L_1, the variation log L_2, and the loop-biased log L_3 (each 100 traces). A = Apply, VC = Verify Credit, VI = Verify Income, I = Interview, Rv = Revise, Ap = Approve, R = Reject.

3 Advantages

Semantic alignment. By moving conformance to stochastic PO languages, it measures process structure rather than recording traces. The example of the paper, where classical EMSC yields 0.689 and PO-EMSC yields 0.955, shows this idea: 45% of the model's probability mass was forced onto two interleavings of one concurrent pair under the totally ordered view, neither of which the log matched exactly; in the PO view the mass sits on a single concurrent PO trace that both interleavings of the log satisfy.

Statistical economy. A single PO trace represents up to $|\mathcal{L}(\rho)|$ total orders, which can be factorial in the width of $\prec$. The applicability study reports 2000 PO traces that collectively represent up to $2.1 \cdot 10^{11}$ classical traces; the same coverage with classical EMSC would require sampling on the order of 10^{20} traces. The bottleneck shifts from sampling to distance computation.

Formal guarantees for loops and uncertainty. The bound lemmas turn a previously heuristic truncation into a verifiable computation: the user knows the true value lies in $[\text{emsc}_{\uparrow}, \text{emsc}_{\downarrow}]$, and can spend more compute on q and r to tighten it. The same lemmas treat log uncertainty (logs with many equal timestamps such as Sepsis at 30%) symmetrically, returning an honest interval rather than a single number that depends on an arbitrary tie-breaking rule.

Decoupling concurrency weights from choice weights. The confusion-free assumption proves that weights on transitions outside any choice do not affect run probabilities, simplifying modelling and matching the practice of translating standard BPMN with exclusive, parallel, and block-structured inclusive gateways into Petri nets.

4 Disadvantages

Worst-case factorial in concurrency width. Counting $|\mathcal{L}(\rho)|$ and computing Δ^{cc} are factorial in the number of pairwise-concurrent events. The implementation defends itself by returning a baseline of 0 whenever $|\mathcal{L}(\rho)| > 10^5$, which loosens the bound. In logs with wide concurrency—exactly those where the PO perspective adds the most value—this fallback weakens the very guarantee that motivates the method.

Slow convergence of emsc-uc**.** The Sepsis experiment leaves a 0.049 gap with $q = r = 1000$, and the BPIC18-4 case shows bound widths up to 0.824 even after 10^6 truncated PO traces. The paper itself flags $\Delta^{uc\text{-worst}}$ as future work. On the very logs where the uncertain interpretation is needed, the technique often returns an interval too wide to discriminate between competing models.

Confusion-freeness and soundness need to be verified manually. While safeness can be guaranteed by the discovery algorithm, there is still a need to verify confusion-freeness and soundness. This is a non-trivial task, especially for large process models. The author mentions that automated checks for these properties still need to be developed, which adds an additional hurdle for practical adoption of the technique.

5 Positioning and Comparison

Among recent stochastic conformance checking (SCC) work, the two approaches closest to the reviewed paper [1]—sharing the (S)LPN/GSPN model formalism and the Earth Movers' (EMSC) lineage—are *abstract-and-compare* (Goulart Rocha, Leemans, and van der Aalst, 2025) [2] and *Enjoy the Silence* (Leemans, Maggi, and Montali, 2024) [3]. Table 2 contrasts the three along input, output, the problem solved, semantics, and underlying assumptions.

All three keep the (S)LPN/GSPN backbone and target SCC, but differ in *what* they compare. The reviewed paper lifts the comparison to partial orders, attacking the *order* dimension—concurrency redundancy and timestamp uncertainty—via optimal transport over PO traces; abstract-and-compare projects both log and model onto subtrace-frequency statistics and then applies any existing measure, buying partial-mismatch robustness and model-class breadth; Enjoy the Silence keeps total orders but, via an absorbing Markov chain, computes their probabilities exactly, introducing an approach where silent-transition and loop/infinite-path mass is dissolved, which the other two consider by removing them during play-out or only by approximation. The contrast is the greatest on loops: PO-EMSC truncates the play-out and returns a provable interval, Enjoy the Silence needs no truncation (but ignores concurrency), and abstract-and-compare avoids enumeration in closed form for livelock-free nets. PO-EMSC's structural requirements are also stronger and, unlike the others, left unverified, and it alone consumes log *timestamps* rather than control flow. The approaches are therefore complementary: Enjoy the Silence's analytic probabilities could replace the truncated play-out to dissolve P3's residual mass, while abstract-and-compare's ab-

	This paper (PO-EMSC)	Abstract-and-compare	Enjoy the Silence
Problems solved	Concurrency redundancy (P2), timestamp uncertainty (P1), loop guarantees (P3)	Robustness to partial mismatch; broad model classes; analytic (no sampling)	Exact trace probability under silent transitions, repeated labels, and loops
Input	LSPN + *timestamped* log	SLPN + log	GSPN/LSP *with silent transitions* (+ log for SCC)
Assumptions	Confusion-freeness, soundness, safeness; usable timestamps	Boundedness, livelock-freeness; choice of subtrace order k	Boundedness; finite-trace semantics; timed/immediate transitions homogeneous
Output	EMSC value, or provable interval $[\text{emsc}_{\uparrow}, \text{emsc}_{\downarrow}]$	Derived SCC measures m^k + conformance diagnostics	Analytic trace / outcome / specification probabilities feeding EMSC-style SCC(uEMSC)
Semantics	Stochastic *partially ordered* language (certain / uncertain); optimal transport over PO traces	Stochastic language of expected subtrace (k-gram) frequencies; then any existing SCC measure	Finite-trace stochastic language via an absorbing Markov chain; totally ordered, order-blind

Table 2. The reviewed PO-EMSC technique against two other techniques

straction provides a robust and scalable lever that the exact-match PO distance lacks.

References

1. Leemans, S.J.J., Brockhoff, T., van der Aalst, W.M.P., Polyvyanyy, A.: Partially ordered stochastic conformance checking. Knowl. Inf. Syst. **67**, 2291–2319 (2025)
2. Goulart Rocha, E., Leemans, S.J.J., van der Aalst, W.M.P.: Abstract-and-compare stochastic conformance checking. Process Sci. **2**, 13 (2025)
3. Leemans, S.J.J., Maggi, F.M., Montali, M.: Enjoy the silence: analysis of stochastic Petri nets with silent transitions. Inf. Syst. **124**, 102383 (2024)